

2023 学年第二学期九年级学业质量调研数学试卷

评分参考

一、选择题：

1 . C ; 2 . D ; 3 . A ; 4 . B ; 5 . C ; 6 . D .

二、填空题：

7 . $xy(y-x)$; 8 . $x=13$; 9 . $x \neq -1$; 10 . $c \geq -\frac{1}{4}$;

11 . $y=(x-3)^2+1$; 12 . $\frac{1}{3}$; 13 . 430 ; 14 . $m \cdot \tan \alpha + m \cdot \tan \beta$;

15 . $a+6b$; 16 . 30 ; 17 . $\frac{1}{4}$. ; 18 . $\sqrt{5}-2 \leq r < 2$ 或 $\sqrt{5} < r \leq \sqrt{5}+2$.

三、解答题：

19 . 解：原式 = $4-1+2\sqrt{5}-\sqrt{5}-3$ (8 分)

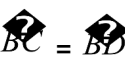
= $\sqrt{5}$ (2 分)

20. 解：由②得 $x + y = 0$ 或 $x - 3y = 0$. (2 分)

原方程组可化为 $\begin{cases} 2x + y = 21, \\ x + y = 0. \end{cases}$ 或 $\begin{cases} 2x + y = 21, \\ x - 3y = 0. \end{cases}$ (4 分)

解得原方程组的解是 $\begin{cases} x_1 = 21, \\ y_1 = -21. \end{cases}$, $\begin{cases} x_2 = 9, \\ y_2 = 3. \end{cases}$ (4 分)

21. 解：(1) 联结 AC .

$\because O$ 为圆心 ,  ,

$\therefore AB \perp CD$, $CE = DE$. (2 分)

$\therefore AC = AD$. (1 分)

又 $\because AD = CD$, $\therefore AC = AD = CD$.

$\therefore \triangle ACD$ 是等边三角形 . (1 分)

$\therefore \angle ADC = 60^\circ$. (1 分)

(2) 联结 OD .

$\because \angle ADC = 60^\circ$, $\therefore \angle A = 30^\circ$. (1 分)

$\because OD = AO$, $\therefore \angle A = \angle ADO = 30^\circ$. (1 分)

$\therefore \angle EDO = 30^\circ$. (1 分)

在 $\text{Rt}\triangle OED$ 中 , $\because DE = \cot 30^\circ \cdot OE$, $OE = 1$, $\therefore DE = \sqrt{3}$. (1 分)

$\therefore AD = CD = 2DE = 2\sqrt{3}$. (1 分)

22. 解：(1) $y = 1500x + 1200(7 - x)$. (2 分)

$$= 300x + 8400 . \dots\dots\dots (1 \text{ 分})$$

$$(2) \because 300x + 8400 \leq 10200 , \therefore x \leq 6 . \dots\dots\dots (1 \text{ 分})$$

$$\because 45x + (7-x)33 \geq 275 , \therefore x \geq 3\frac{2}{3} . \dots\dots\dots (1 \text{ 分})$$

$$\because x \text{ 为整数} , \therefore x \text{ 可取 } 4 , 5 , 6 . \dots\dots\dots (1 \text{ 分})$$

$$\therefore \text{一共有 } 3 \text{ 种租车方案} . \dots\dots\dots (1 \text{ 分})$$

$$(3) \because \text{一次函数 } y = 300x + 8400 , k = 300 > 0 , \therefore y \text{ 的值随着 } x \text{ 的值增大而增大} .$$

$$\because x \text{ 可取 } 4 , 5 , 6 , \therefore \text{当 } x = 4 \text{ 时} , y \text{ 的值最小} . \dots\dots\dots (1 \text{ 分})$$

$$\text{把 } x = 4 \text{ 代入 } y = 300x + 8400 , \text{得 } y = 9600 . \dots\dots\dots (1 \text{ 分})$$

$$\therefore \text{选择租用甲种型号的客车 } 4 \text{ 辆} , \text{乙种型号的客车 } 3 \text{ 辆最省钱} ; \dots\dots\dots (1 \text{ 分})$$

$$\text{此时租车的总费用为 } 9600 \text{ 元} .$$

$$23 . \text{解} : (1) \because AD \parallel BC ,$$

$$\therefore \angle DAB + \angle CBA = 180^\circ , \angle DAE = \angle ACB . \dots\dots\dots (2 \text{ 分})$$

$$\text{又} \because \angle DAB = \angle BCD ,$$

$$\therefore \angle BCD + \angle CBA = 180^\circ .$$

$$\therefore AB \parallel CD . \dots\dots\dots (1 \text{ 分})$$

$$\therefore \text{四边形 } ABCD \text{ 为平行四边形} . \dots\dots\dots (1 \text{ 分})$$

$$\because \angle DEC = \angle DAB , \angle DEC = \angle ADE + \angle DAE , \angle DAB = \angle BAC + \angle DAE .$$

$$\therefore \angle ADE = \angle BAC .$$

$$\because EA = ED, \therefore \angle ADE = \angle DAE.$$

$$\therefore \angle DAE = \angle BAC = \angle ACB.$$

$$\therefore BA = BC. \dots\dots\dots (1 \text{ 分})$$

$$\therefore \text{四边形 } ABCD \text{ 是菱形}. \dots\dots\dots (1 \text{ 分})$$

(2) $\because$ 四边形 $ABCD$ 是菱形,

$$\therefore DC = AD = BC = AB. \dots\dots\dots (1 \text{ 分})$$

$$\therefore AD \parallel BC, \therefore \frac{AD}{GB} = \frac{AF}{FB}. \dots\dots\dots (1 \text{ 分})$$

$$\therefore GB = BC, \therefore GB = AD. \therefore AF = FB.$$

$$\therefore AF = \frac{1}{2} AB = \frac{1}{2} AD. \dots\dots\dots (1 \text{ 分})$$

$$\therefore AB \parallel DC, \therefore \angle AFE = \angle GDC.$$

$$\therefore \angle DEC = \angle AEF, \angle DEC = \angle DCB.$$

$$\therefore \angle AEF = \angle DCB.$$

$$\therefore \triangle AEF \sim \triangle GCD. \dots\dots\dots (1 \text{ 分})$$

$$\therefore \frac{EF}{CD} = \frac{AF}{GD}. \dots\dots\dots (1 \text{ 分})$$

$$\therefore AF \cdot CD = EF \cdot GD.$$

$$\therefore \frac{1}{2} AD \cdot AD = EF \cdot GD.$$

$$\therefore AD^2 = 2EF \cdot GD. \dots\dots\dots (1 \text{ 分})$$

24. 解: (1) $\because$ 抛物线 $y = ax^2 + bx - 3$ 经过点 $A(-3, 0)$ 和 $B(1, 0)$,

$$\therefore \begin{cases} 9a - 3b - 3 = 0 \\ a + b - 3 = 0. \end{cases} \dots\dots\dots (2 \text{ 分}) \text{ 解得: } \begin{cases} a = 1, \\ b = 2. \end{cases}$$

$$\therefore \text{抛物线的解析式为 } y = x^2 + 2x - 3. \dots\dots\dots (1 \text{ 分})$$

$$\text{当 } x = 0 \text{ 时, } y = -3. \therefore \text{点 } C \text{ 的坐标为 } (0, -3). \dots\dots\dots (1 \text{ 分})$$

(2) 过点 C 作 $CH \perp DG$ ，垂足为点 H 。

设点 D 为 $(m, 0)$ ($m < 0$)。

$$\therefore G(m, m^2 + 2m - 3) \quad \dots\dots\dots (1 \text{ 分})$$

$$\therefore H(m, -3) \quad .$$

$$\therefore AD = 3 + m, DG = -m^2 - 2m + 3 \quad \therefore HC = -m, HG = -m^2 - 2m \quad .$$

$$\therefore \angle DGA = \angle DGC, \therefore \tan \angle DGA = \tan \angle DGC \quad .$$

$$\therefore \frac{AD}{DG} = \frac{HC}{HG} \quad \therefore \frac{m+3}{-m^2-2m+3} = \frac{-m}{-m^2-2m} \quad \dots\dots\dots (1 \text{ 分})$$

$$\text{解得 } m = -\frac{1}{2} \text{ 或 } m = -3 \text{ (舍去)} \quad .$$

$$\therefore \text{点 } D \text{ 的坐标 } \left(-\frac{1}{2}, 0\right) \quad .$$

$$\therefore \text{点 } G \text{ 的坐标 } \left(-\frac{1}{2}, -3\frac{3}{4}\right) \quad \dots\dots\dots (1 \text{ 分})$$

$$\therefore S_{\triangle AGC} = S_{\text{梯形}DOCG} + S_{\triangle ADG} - S_{\triangle AOC} = \frac{15}{8} \quad \dots\dots\dots (1 \text{ 分})$$

(3) 设 BP 与 DC 相交于点 Q 。过点 B 、 Q 作 $BM \perp DQ$ 、 $QN \perp DO$ ，垂足分别为点 M 、 N 。

$$\therefore \text{点 } D \text{ 的坐标 } (-1, 0), \text{ 点 } C \text{ 的坐标为 } (0, -3),$$

$$\therefore \tan \angle ODC = \frac{OC}{OD} = 3 \quad \therefore DM = \frac{1}{5}\sqrt{10}, \quad BM = \frac{3}{5}\sqrt{10} \quad .$$

$$\therefore \angle DCB + \angle PBC = 45^\circ, \therefore \angle DQB = 45^\circ \quad .$$

$$\therefore BM = QM = \frac{3}{5}\sqrt{10} \quad \therefore QD = \frac{4}{5}\sqrt{10} \quad \dots\dots\dots (1 \text{ 分})$$

$$\therefore DN = \frac{4}{5}, \quad QN = \frac{12}{5} \quad .$$

∴点 Q 的坐标 $(-\frac{1}{5}, -\frac{12}{5})$ (1 分)

可得直线 BQ 的解析式为 : $y = 2x - 2$ (1 分)

设点 P 的坐标为 $(n, 2n - 2)$.

将 $(n, 2n - 2)$ 代入 $y = x^2 + 2x - 3$,

得 $n = -1$ 或 $n = 1$ (舍去) ∴点 P 的坐标 $(-1, -4)$ (1 分)

25 . 解 : (1)

①∵ $AB = AC$, $\angle ABC = \alpha$,

∴ $\angle ACB = \angle ABC = \alpha$ (1 分)

∵ $CB = CD$, ∴ $\angle CDB = \angle DBC$.

∵ $CE = CD$, ∴ $\angle CED = \angle CDE$.

∵ $\angle CED + \angle CDE + \angle CDB + \angle DBC + \angle ACB = 360^\circ$,

∴ $2(\angle CDE + \angle CDB) + \alpha = 360^\circ$ (1 分)

∴ $\angle CDE + \angle CDB = 180^\circ - \frac{1}{2}\alpha$ (1 分)

∴ $\angle BDF = 180^\circ - (180^\circ - \frac{1}{2}\alpha) = \frac{1}{2}\alpha$ (1 分)

②∵ $\angle DBC = \alpha$, $\angle BDF = \frac{1}{2}\alpha$, ∴ $\angle F = \angle CDB - \angle BDF = \frac{1}{2}\alpha$.

∴ $\angle F = \angle BDF$. ∴ $BF = BD = 1$ (1 分)

过点 C 作 $CH \perp DB$, 垂足为点 H .

$$\because CB = CD, \therefore BH = \frac{1}{2}BD = \frac{1}{2} \dots\dots\dots (1 \text{ 分})$$

$$\because \angle ACB = \angle ABC = \angle BDC, \therefore \triangle CBD \sim \triangle ABC \dots\dots\dots (1 \text{ 分})$$

$$\therefore \frac{BC}{AB} = \frac{BD}{BC} \therefore CB = \sqrt{2} \dots\dots\dots (1 \text{ 分})$$

$$\therefore \cos \angle ABC = \frac{BH}{BC} = \frac{\sqrt{2}}{4} \dots\dots\dots (1 \text{ 分})$$

(2) 设 AB 与 MN 交于点 K . 设 $\angle ABC = \alpha$, $BC = x$.

$$\because AB = AC, CB = CD, \therefore \angle ACB = \angle ABC = \angle BDC .$$

$$\because \triangle CBD \sim \triangle ABC \therefore \frac{BC}{AB} = \frac{BD}{BC} \therefore BD = \frac{1}{2}x^2 \dots\dots\dots (1 \text{ 分})$$

$$\because \angle ACB = \angle ABC = \angle BDC = \alpha ,$$

$$\therefore \angle DCB = 180^\circ - 2\alpha .$$

$$\therefore \angle ACD = \angle DCB - \angle ACB = 180^\circ - 3\alpha ,$$

$$\because CE = CD, \therefore \angle CDE = \angle CED = \frac{180 - (180 - 3\alpha)}{2} = \frac{3}{2}\alpha .$$

$$\therefore \angle BDF = \angle CDE - \angle CDB = \frac{1}{2}\alpha .$$

$$\angle F = \angle CDB - \angle BDF = \frac{1}{2}\alpha \therefore BF = BD = \frac{1}{2}x^2 \dots\dots\dots (1 \text{ 分})$$

$\therefore$

取 BF 的中点 G , 联结 NG , $\therefore NG$ 为 $\triangle BDF$ 中位线 .

$$\therefore NG = \frac{1}{2} BD = \frac{1}{4} x^2 .$$

$$\therefore BG = \frac{1}{2} BD = \frac{1}{4} x^2 . \dots\dots\dots$$

(1 分)

$\therefore MN \parallel CE$, $AB = 2$, M 为 BC 中点 ,

$$\therefore \frac{BM}{BC} = \frac{BK}{AB} = \frac{1}{2} \therefore BK = 1 , BM = \frac{1}{2} x .$$

$$\begin{array}{l} \therefore \frac{BM}{MG} = \frac{BK}{NG} \quad \therefore \frac{\frac{x}{2}}{\frac{x}{2} + \frac{x^2}{4}} = \frac{1}{\frac{x^2}{4}} \dots\dots\dots (1 \text{ 分}) \\ \therefore \end{array}$$

$$\therefore x = 1 + \sqrt{5} \text{ 或 } x = 1 - \sqrt{5} \text{ (舍去)} .$$

$$\therefore CB \text{ 的长 } 1 + \sqrt{5} . \dots\dots\dots (1 \text{ 分})$$